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Review article

Interpreting NUMTs in forensic genetics: Seeing the forest for the trees

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ABSTRACT

Nuclear mitochondrial DNA (mtDNA) segments (NUMTs) were discovered shortly after sequencing the first human mitochondrial genome. They have earlier been considered to represent archaic elements of ancient insertion events, but modern sequencing technologies and growing databases of mtDNA and NUMT sequences confirm that they are abundant and some of them phylogenetically young. Here, we build upon mtDNA/NUMT review articles published in the mid 2010 s and focus on the distinction of NUMTs and other artefacts that can be observed in aligned sequence reads, such as mixtures (contamination), point heteroplasmy, sequencing error and cytosine deamination. We show practical examples of the effect of the mtDNA enrichment method on the representation of NUMTs in the mapped sequence data and discuss methods to bioinformatically filter NUMTs from mtDNA reads.

1. Introduction

The analysis of mitochondrial DNA (mtDNA) has a long tradition in forensic genetics. It was one of the first markers analyzed by direct sequencing and still is one of the most differentiating single genetic markers available to forensic genetic practice [1–4]. We refer to previous reviews that describe the benefits and limitations of mtDNA in forensic genetics in detail, e.g., [5]. In brief, mtDNA is more abundant in (human) cells than nuclear DNA (nDNA) [6], which is why mtDNA-targeted assays can provide useful results in forensic samples that do not contain enough nDNA for successful DNA profiling. Therefore, mtDNA analysis is often considered as the last resort of hope in those cases, where nDNA analysis has failed to provide meaningful results. However, mtDNA typing is also employed in situations where enough high quality nDNA would be available, because it provides a strong and reliable signal across many generations that could not be achieved with recombining nDNA markers (e.g., [7]).

The mitochondrial genome (mitogenome) is one of the first genetic markers to be analyzed using Next Generation Massively Parallel Sequencing (MPS) techniques [8] in the forensic context. The fact that mtDNA has always been sequenced and reported as a haplotype relative to a reference sequence, the rCRS [9], made the transition from electrophoretic Sanger-based methods to MPS relatively easy. This is in contrast to other forensic genetic markers that were not traditionally sequenced, such as STRs, leading to fundamental sequence analysis and nomenclature issues that are currently under debate and therefore postpone the process of wider implementation [10,11]. The transition of mtDNA analysis from Sanger to MPS brought several advantages to forensic applications that have been summarized (e.g., [5]) and are discussed in more detail in this review.

MtDNA is maternally and unilaterally inherited in humans [12]. This dogma has been repeatedly subject to debate in the past decades; however, most of the assertions were based on flawed data, used unjustified claims or suffered inappropriate methodological approaches (see a summary in [13]). As we outline in this review, paternal contribution to mtDNA inheritance has also been suggested recently using MPS based techniques [14]. The conclusions in Luo et al.2018 were challenged [15–17], and an alternative explanation for the observations was offered [16] that is the focus of this review – NUMTs – nuclear elements of mtDNA.

2. A brief historical sketch of NUMTs

The circular human mitogenome was for the first time sequenced in 1981 and has become known as the Cambridge Reference Sequence (CRS) [18]. The same mtDNA was analyzed again with modern Sanger

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sequencing technology 18 years later and corrected for earlier sequencing errors [9]. This revised version of the CRS, also known as the rCRS, has become the reference sequence relative to which all human mtDNA sequences have been reported.

The human mitogenome contains the 1.1 kbp non-coding Control Region (CR), which has traditionally been investigated in forensic genetics and other fields of research, and the 15.4 kbp coding region (codR) that harbors 13 genes, 22 tRNAs and 2 rRNAs [9,18]. This minimal repertoire of genetic information present in the mitogenome requires an active import of proteins, enzymes and co-factors from the nucleus into the mitochondrion to enable the organelle's functionality. This, as well as the similarity of mtDNA with bacterial genomes, has contributed to the formulation of the Serial Endosymbiosis Theory, according to which mitochondria derive from Proteobacteria [19]. In this symbiotic association mitochondrial genes have progressively been transferred from mtDNA to the nuclear genome, which has been documented for humans and other species already in 1983 ([20] and references therein). The term 'nuclear mtDNA segment', short *NUMT*, was coined by Lopez et al.[21], who reported the transposition of a 7.9 kbp mtDNA fragment into nDNA of the domestic cat. The mechanism by which NUMTs are entering the nuclear genome are not fully understood, but studies in yeast demonstrated that NUMTs integrate themselves through a non-homologous end-joining mechanism during double-strand break repair [22,23]. This same mechanism was indicated by Jensen-Seaman and colleagues based on NUMT insertion patterns in the human and chimpanzee genomes [24]. With the emergence of MPS-based technologies, a wide variety of species' genomes have been sequenced, which has shown that almost every genome of any species contains NUMTs [25].

High-throughput sequencing data from different human populations have further substantially increased our knowledge on NUMT location and evolution. For example, a study on 1000 Genomes Project data discovered 141 new NUMTs ranging from 39 to 16,106 bp in size in addition to the established catalogue of ~755 NUMTs [26]. Prior to this study, only a small fraction of NUMTs were believed to represent recent insertions, as the majority was found in every individual. Phylogenetic analyses of the observed nuclear inserts in [26], including new ones in the non-coding control region, however, suggest that most of the NUMTs have been inserted within the past 100.000 years. Furthermore, over 50% of the NUMTs identified in [26] showed very low frequencies across the tested samples, which indicates that these NUMTs are young and the process of NUMT insertions into the nuclear genome is still ongoing [26,27]. NUMTs initially arise by mitochondrial insertion events, but they continue to evolve within the nucleus by duplication. The duplication rate in humans was estimated to be $2.2 * 10^{-9}$/year/-NUMT, which is similar to the substitution rate of Single Nucleotide Polymorphisms (SNPs) [28]. Additional NUMT sequences have since been published [25,29]. These, along with the NUMTs presented in [26], have been compiled by Woerner et al. [30] in a database of 1090 NUMT sequences (Table 1). The registered NUMTs range from 13 to 18,649 bp with three NUMTs exceeding the full length of a mitogenome (>16, 569 bp).

Table 1
NUMT fragment length in base pairs (bp), based on the NUMT sequence database compiled by Woerner et al. [30].

NUMT Fragment Length (bp)	Count	Proportion
13–99	232	21.3%
100–499	665	61.0%
500–999	63	5.8%
1,0000–4999	89	8.2%
5000–9999	32	2.9%
10,000–18,649	9	0.8%
Total	1090	100.0%

3. Mega-NUMTs

Balciuniene and Balciunas [31] speculated that NUMTs can also be present as multi-copy full mitogenome inserts, a phenomenon that the authors termed mega-NUMT [31]. In fact, physical evidence for a mega-NUMT consisting of approx. 50 copies/nucleus was recently provided in eight of 11 healthy, maternally related family members across three generations [32]. The authors confirmed the presence of the mega-NUMT by applying Sanger and MPS mtDNA sequencing approaches on DNA extracted from various tissues (e.g., blood, saliva) and clones developed from these tissues. The mega-NUMT was evident as a full mitogenome of haplogroup (hg) U4c1 status in a mixture with the genuine mtDNA of the maternal lineage (hg V1). The hg U4c1 insert was only observed in cells containing nDNA, while sequencing mtDNA from acellular sources, such as hair shafts and thrombocytes, only yielded the V1 mitotype. The authors examined ρ^0 cells (void of mtDNA) generated from a skin biopsy of one affected family member and observed only the U4c1 mitotype, corroborating that this sequence does not derive from mitochondria. Also, full mitogenome MPS from rootless hair shafts, which are known to harbor only few to no detectable copies of high molecular weight nDNA, exclusively resulted in the V1 mitotype in the affected family members, thus excluding the possibility that the U4c1 mitotype originated in the mitochondria. It has been demonstrated that rootless hair shafts do contain nDNA, even in much higher molarity than mtDNA, however not as readily amplifiable as mtDNA by PCR due to its high fragmentation [33]. Finally, FISH experiments located the mega-NUMT on Chr. 14 [32].

Mega-NUMTs were also reported from an in-silico analysis in seven of 11,035 trios occurring at minor allele frequencies (MAF) of 5–25% on Chromosomes 3, 7, 12, 13 and 17 [34]. Only fathers were observed to pass the mega-NUMT to the next generation in that study, which may be related to the authors' sampling strategy for NUMT detection (see Fig. 3 in [34]). Finally, a strikingly similar constellation to the mega-NUMT observed in [32] was reported for three pedigrees that the authors interpreted as 'biparental inheritance of mtDNA' [14]. The authors were deducing the "paternal" components in the observed mtDNA mixtures and assigned them to the fathers of the respective pedigree members without confirming (sequencing the fathers mtDNAs) or challenging their assumptions experimentally [16]. Our current knowledge suggests that mega-NUMTs were more likely the source for the observed mixtures.

All these observations strongly suggest that mega-NUMTs do exist in various forms and on different chromosomes, however they are probably rare in the general population. Mega-NUMTs at appreciable MAFs (> 10%) could hamper interpretation of forensic and medical genetic sequencing data as their co-amplification with the genuine mtDNA cannot be avoided. Only tissues that naturally lack intact nDNA, such as hair shafts, or those that lack mtDNA, such as ρ^0 cells, would then serve as evidence for testing the presence of mega-NUMTs with PCR-based approaches. It can be expected that mega-NUMTs would still appear in hair shafts using capture or whole genome sequencing (WGS) [33].

4. NUMTs in forensic applications

Early studies that specifically investigated NUMTs in human forensic samples and even tried to provoke their co-amplification with the genuine mtDNA were largely unsuccessful (e.g., [35]). The authors used PCR primers that were directed to either the mtDNA or the NUMT component, and varied annealing temperatures. They concluded that only tissue depleted of mtDNA, such as semen after differential lysis, would pose the risk of showing results with increased NUMT contributions. Instead, NUMTs were identified in feline samples despite the scarce information available on the cat's genome [36]. Sanger-based sequencing was used in those early studies that is generally less sensitive to the detection of low-level mixture contributions and also, the number of known human NUMTs was relatively small at that time. Later,

MPS studies clearly demonstrated that NUMT co-amplification is a common phenomenon (e.g., [37,38]), affecting even CR sequencing [37]. MPS allows for a wider range of applications including both long amplicon and multiplex PCR-based mini amplicon primer-tiling approaches (e.g. [38–41]), primer extension protocols as used in the QIAseq chemistry [42], as well as capture-based enrichment methods [43–45] that are not amenable to Sanger-based sequencing in a routine environment. The technology has meanwhile matured and provides a comparably cost-effective and less labor-intensive solution compared to Sanger-based techniques, particularly when many samples or full mitogenomes are analyzed [46].

The forensic application of mitogenome MPS, which expanded the targeted portion from the CR to the entire mitogenome, meanwhile increased the abundance of observed and reported NUMTs (e.g., [42,47–49]. This is in part due to the enhanced sensitivity gained with MPS, which allows for the detection of low frequency variants in mtDNA sequence alignments, such as those originating from NUMT reads. Yet, the main reason for picking up NUMTs in MPS-based studies is that most applications target hundreds of mini amplicons/ mtDNA fragments using primers homologous to both mtDNA and NUMTs. Mini amplicon workflows are relatively simple to perform (e.g., [50]) and automate (e.g., [49]) in the laboratory; they are also increasing the success rate of mtDNA analysis in even heavily degraded mtDNA that is typically present in old and/or environmentally challenged specimens [39,44,51–53]. However, they are also more likely to enrich NUMT fragments than long-range mtDNA workflows do [42,48,49,54].

To illustrate the effect of the enrichment method on NUMT interference in mtDNA sequence alignments, we present an example from a reference-type blood sample that was sequenced using different mitogenome enrichment approaches: WGS with no targeted enrichment (also called shotgun sequencing), hybridization capture, long amplicon and mini amplicon workflows. Fig. 1 shows the rCRS-aligned reads from nucleotide positions ~650–850 in the coding region, a stretch of mtDNA that is prone to NUMT interference [48,49]. This particular haplotype exhibits a variant at np 750 (A750G), which constitutes a signature mutation for hg H2a2 (Phylotree b.17 [55]) and is thus present in almost every haplotype that is reported relative to the rCRS (hg H2a2a1). This variant is most clearly shown in the long amplicon tract. Aside from A750G, this haplotype did not harbor any variants and should therefore not produce any other differences to the rCRS in the mtDNA reads. The mtDNA profile can most clearly be seen in the long amplicon data that is lacking NUMT signal, although sporadic sequence variants due to PCR and/or sequencing error were present (Fig. 1). In the three remaining alignment tracks (WGS, mini amplicon, and hybridization capture), an A can be observed at 750 due to a NUMT along with additional NUMT-derived variants in the reads. When multiple variants can be observed along a single sequence read, they are deemed “in-phase” or “phased”, having originated from the same DNA strand. Reads with phased variants can originate from both mtDNA and NUMTs. Yet in this particular example, the high number of phased variants in a short stretch of coding region sequence helps to identify them as NUMTs. The NUMT reads appear to be most prevalent in the WGS and mini amplicon data (estimated at ~9% and ~19%, respectively), followed by hybridization capture (~3%). Since the same sequencing method and the same alignment parameters were used across all enrichment conditions of the same DNA sample, the varying levels of NUMT interference stem from



Fig. 1. Reads aligned to the rCRS after several different enrichment approaches that were applied to a single reference sample (blood): A) whole genome sequencing (shotgun), B) mini amplicon, C) hybridization capture, and D) long amplicon. The green horizontal lines represent forward sequence reads, the red horizontal lines represent reverse sequence reads. Variants from the rCRS are indicated by color: red=adenine (A), yellow=guanine (G), blue=cytosine (C), and green=thymine (T). The variant-containing reads are pinned at the top of the alignments. This individual exhibits a mtDNA variant at nucleotide position 750 (A750G), which is most clearly viewed in the long amplicon tract. The reads with multiple variants from the rCRS originate from NUMTs. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

the enrichment approach. Specifically, long amplicon enrichment produces the fewest NUMT reads, followed by hybridization capture, mini amplicon, and then WGS. Thus, although mini amplicon enrichment may be preferable in forensic applications due to its ability to handle degraded DNA, it produces more NUMT reads in the mtDNA sequence data than other enrichment methods.

As demonstrated above, one of the most important features of MPS is the ability to understand the phasing of variants in the parsed sequence data, which is limited in electropherograms produced from Sanger sequencing. With MPS, one can generally discern linked variants residing on the same mtDNA molecule, from unlinked variants that originate from separate DNA molecules with the caveat of artefactual arrangements that can be caused by jumping PCR. Thus, although mitogenome MPS results in NUMTs being commonly observed, it also offers new solutions to diagnose NUMTs in the sequence data. This is especially important when combined with other types of low frequency (generally <10%) variants, such as those arising from mixtures/contamination, cytosine deamination, error, and true heteroplasmy (Table 2). To an untrained analyst, low frequency variants may be difficult to discern. Yet, each of them has certain properties that help with its diagnosis when assessing threshold-based variant calls. NUMTs are likely to be observed at known NUMT locations [29,30,42,47–49,56], whereas point heteroplasmy – particularly in the codR – [57–61] and stochastic error occur at random locations across the mitogenome [62–64]. Furthermore, point heteroplasmy is not observed in a majority of mitogenome haplotypes when applying variant detection thresholds realistic to forensic samples, i.e., a minimum of 5–10% [57–61]. NUMTs (including mega-NUMTs [16,31,32,34]) and point heteroplasmy are reproducible across PCR events from the same DNA sample (or library events, if performing WGS or capture), whereas stochastic error is not reproducible. Cytosine deamination may be more readily identified than point heteroplasmy, because it causes specific base substitutions (C → T and G → A) at the ends of natural (non-amplified), degraded DNA fragments. Variants caused by cytosine deamination have been observed in forensic samples [65–69], and these random variants are not reproducible across PCR or library preparation events. Mixtures may be reproducible from the same DNA sample, yet they can be readily identified when the mixed samples have differing haplogroups, producing low frequency variants at haplogroup-diagnostic sites [44,63,70–75]. Like NUMTs, mixtures can also produce phased variants depending on the haplotypes that were mixed. What may distinguish commonly observed NUMTs – not mega-NUMTs - from mixtures is the location of the variants, as such previously characterized NUMTs exhibit variants at specific locations where NUMTs are prone to align to the rCRS. Sequencing error is another source of low frequency variants [73,76], and this type of error can be flagged by strand imbalance, in which one read direction (forward or reverse) contains a variant that is not found in the opposite direction. Sequencing error may introduce variants with reduced base quality; however, reduced sequence quality is not a diagnostic characteristic of NUMTs.

Typically, NUMTs are present as single fragments or in small copy number in the nuclear genome. This is why mtDNA analysis in general, and particularly in tissues of forensic interest (e.g., rootless hair shafts, degraded bone/teeth), does not yield NUMT signals above background noise. However, NUMTs may appear frequently in reference-type samples assayed with mini amplicon enrichment techniques [42,47,49]. One method to address NUMT co-amplification in reference samples and relatively high-quality forensic samples is to control (i.e., reduce) the quantity of genomic DNA prior to PCR [49,70,77]. Since mtDNA averages several hundred copies per cell compared to the two copies of nDNA [6,78], it is possible to dilute the DNA so that only several dozen copies of mtDNA remain and zero copies of nDNA end up in the PCR [77]. This approach effectively "dilutes out" the NUMTs entirely, leading to NUMT-free mtDNA sequences. This is why quantitation of the mtDNA and the nDNA in a sample to direct the input quantity for mitogenome PCR may be important in mini amplicon laboratory workflows (e.g., [79,80]), or those involving high-quality DNA followed by WGS or hybridization capture. Validation studies have demonstrated that mini amplicon strategies require mtDNA amounts as little as the equivalent of a single cell or even less as template to generate reliable sequences [81], and, as little as 23 copies of mtDNA to generate a full mitogenome sequence [47]. This indicates that smaller amounts of template mtDNA are sufficient for mitogenome sequencing. The application of upfront DNA (nuclear and mt) quantitation and DNA dilution can therefore help to reduce the occurrence of NUMTs substantially. However, the dilution approach may not be practical for all mtDNA sequencing applications. For example, in mixture scenarios, the proportions of mtDNA and nDNA from the various contributors may be low, unpredictable, and difficult to control. A dilution of a low-level mixture could possibly eliminate the signal from a low-level contributor. Furthermore, due to mtDNA copy number variation in human tissues [82], the propensity for NUMT enrichment may be related to the source tissue. Those tissues with higher mtDNA copy number, such as buccal samples, will produce fewer NUMTs than those with lower mtDNA copy number, such as blood, given that cellular nDNA copies are constant. Semen samples can be a challenging source of DNA in this respect, as the mtDNA to nDNA ratio can be significantly reduced. This can particularly concern samples with enriched sperm heads (e.g., the pellet of a differential lysis fraction) or older/decomposed stains that lack tails and connecting piece of the flagellum.

5. Bioinformatic filtering of NUMTs

Since it may not be possible to fully avoid NUMTs in mitogenome sequencing, especially considering mega-NUMTS, bioinformatic handling of NUMTs constitutes a critical aspect of data analysis. There are two simple approaches to minimize NUMT interference in a single step, although they have also been shown to reduce mtDNA read depth. The first is to increase the stringency of the mapping parameters to preclude the divergent NUMT sequences from aligning to the rCRS [42]. This can be done by increasing the similarity fraction or the length fraction of aligned bases, or both, to require a higher proportion of matched DNA sequence between the MPS reads and the rCRS. Fig. 2 shows the effect of increasing the mapping stringency when aligning

Table 2
Types of low frequency variants observed in mitogenome sequence alignments and their defining characteristics.

Low Frequency Variant Type	Base Change	Location	Indicator (s)	Reproducible in same sample?
NUMT	Any	Known NUMT locations	Phasing of variants; Low mtDNA: nDNA ratio	Yes
Mega-NUMT	Any	Phylogenetically plausible	Reduced variant frequency overall	Yes
Point Heteroplasmy	Any	Random (hotspots in CR)	Observed in <30% of mitogenomes; When observed, typically 1 but up to 3 per haplotype	Yes
Mixture	Any	Phylogenetically plausible	Reduced variant frequency overall	Depends on ratio
Cytosine Deamination	C→T/ G→A	5' and 3' ends of unmodified DNA fragments; throughout PCR-amplified DNA fragments	Degraded DNA	No
Stochastic Error	Any	Random	Poor amplification yield	No
Sequencing Error	Any	Homopolymer regions; Random	Strand imbalance; Reduced variant quality	Yes



Fig. 2. Whole genome sequencing data mapped to the rCRS using default / relaxed (A) and stringent (B) alignment parameters.

reads to the rCRS after WGS. Clearly, there are fewer NUMT reads in the Fig. 2B alignment utilizing stringent parameters (<1%), compared to the Fig. 2A alignment with default / relaxed mapping (~9%). The reduction in NUMT reads can also correspond with a reduction in mtDNA reads (in this case by ~15%), especially when the haplotype is phylogenetically distant from the rCRS or when imbalanced mixtures of dissimilar haplotypes are present [42].

Another method is to map the sequence data to the entire human reference genome including the rCRS [54,83], to allow NUMTs to map to chromosomal DNA. Fig. 3 shows the rCRS-aligned reads after mapping the same WGS data presented in Fig. 2A to the entire human reference genome (hg38). When compared to Fig. 2A, the aligned reads after hg38 mapping shown in Fig. 3 reveal fewer NUMT inclusions. However, mtDNA reads may also be (falsely) aligned to the chromosomal DNA, thereby reducing the read depth on the reads aligned to the rCRS (by ~13% in this example using default mapping) [83,84]. A reduction in mtDNA read depth can significantly impact heteroplasmy calls, including those of pathogenic variants, and may lead to reference bias favoring the reference allele [84].

To address reference bias, Ring et al. [42] utilized consensus sequence re-mapping to minimize NUMT-associated variants in mini amplicon and primer extension data. The consensus sequence re-mapping first aligns reads to the rCRS to produce a consensus sequence (or major haplotype), using the base with the highest frequency at each nucleotide position. The rCRS-aligned reads are then extracted and re-mapped to the consensus sequence using stringent mapping parameters. This method minimizes NUMT inclusions while maximizing the percent of aligned mtDNA reads, independent of haplogroup or phylogenetic relationship to the rCRS. A similar workflow with re-mapping to the sample's haplogroup was also suggested by

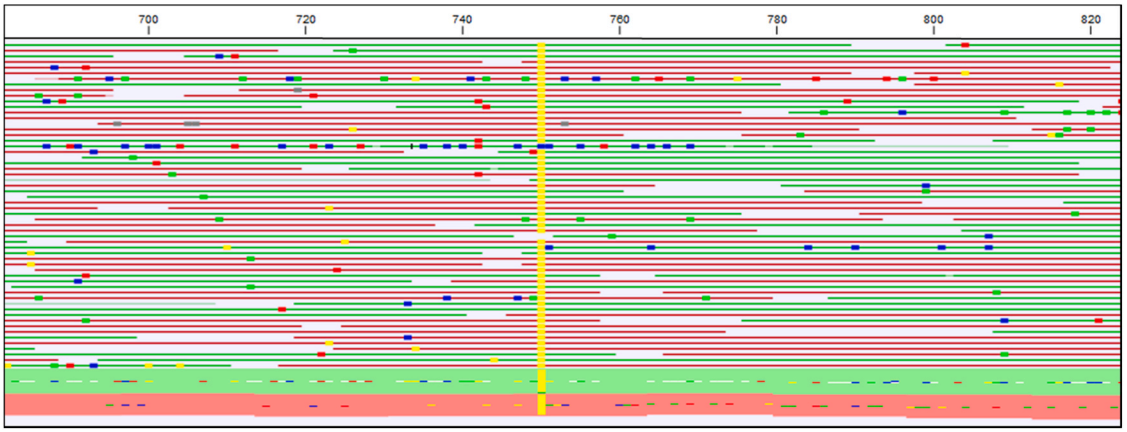


Fig. 3. Reads aligned to the rCRS after whole genome sequencing data are mapped to the entire human reference genome (hg38) including the rCRS.

Maude et al. [84]. Alternatively, an alignment-free approach to NUMT filtering was recently described by Smart et al. [29]. This method employs phylogenetic networks to reveal sequences divergent from the major haplotype in MPS reads, known as putative NUMTs, which are then filtered. The drawback to the consensus re-mapping and network discovery approaches is that authentic mtDNA sequence variants originating from mixtures, or even from heteroplasmy involving multiple nucleotide variants, could be filtered out of the profile. A haplotype-independent method of NUMT filtering was published by Woerner et al. in 2020 [30], and this method is suitable for all sample types including mixtures and heteroplasmic mtDNA. The approach relies on a NUMT sequence database, as mentioned above, as the basis for Woerner et al.'s "Remove the Numts!" (RtN!) application [85]. Rtn! remaps reads to both a human haplotype database as well as the NUMT sequence database. Based on these results, the reads are categorized and the "NUMT-like" reads are filtered from the sequence data. When the authors applied RtN! to mini amplicon MPS data, they found that ~57% of false-positive variants, largely NUMTs, were removed. The authors caution that true mtDNA variation may also be lost when applying an automatic NUMT filter [30]. Therefore, interpretation of the NUMT filtering results is critical to the analysis of a forensic sample.

The effectiveness of bioinformatic NUMT filtering decreases as DNA sequence length decreases [84]. Variant phasing can be lost when NUMTs are fractioned into small segments (e.g., 75–100 bp), and this may be a consequence of DNA shearing prior to capture or WGS, truncated sequences from short read lengths (e.g., [86]), or PCR amplification of small mtDNA segments. As explained above, forensic mitogenome MPS routinely requires targeted PCR enrichment of mini amplicons from degraded and low template DNA fragments. This may limit the option of increasing DNA fragment size and read length in some forensic samples in order to preserve variant phasing along NUMT sequences.

6. Conclusions and forensic relevance

The history of NUMTs has some mystic elements, not for nothing were they labeled with symbolic terms, such as 'pseudogenes' or molecular 'poltergeists' [27]. Their discovery was closely connected to the developments in sequencing technologies. NUMTs were initially believed to occur only rarely based on Sanger sequencing data. To the contrary, MPS data suggest that they are not only common and present in every individual, a large portion of NUMTs is in fact phylogenetically young and the process of mtDNA fragments being inserted into nDNA is still ongoing. While the fragment size distribution of NUMTs tends to be biased towards short elements, large inserts up to repetitive full mitogenome NUMTs (mega-NUMTs) have also been described, albeit less frequently. The latter may pose a particular problem to analytical workflows, as the apparent mixture of the genuine mtDNA with the NUMT contribution can only be avoided with complex technical procedures that are not practical for forensic workflows [87]. Only the analysis of dedicated tissues, either void of high molecular nDNA, such as hair shafts, or lacking mtDNA, such as ρ^0 cells, would disentangle the mtDNA from the mega-NUMTs by either providing only the genuine mtDNA sequence or only the NUMT sequence, respectively.

The presence of NUMTs in rCRS-aligned sequence data is strongly dependent on the mitogenome enrichment method. We show in this review that the same DNA extract can exhibit significantly different NUMT contributions, with Whole Genome Sequencing (shotgun) providing the largest NUMT contents and long-range amplicon the smallest. Apparently and in line with the size distribution of NUMTs, mini amplicon and hybridization capture enrichment methods are prone to NUMT co-amplification, putting the forensic application of mtDNA analysis in a dilemma. That is, mini amplicon (and hybridization capture) enrichment methods are requisite for challenged forensic samples as the success rate of providing useful results increases with decreasing fragment size. However, the resulting sequencing data from reference-type samples processed with these same small fragment techniques are more prone to be contaminated by NUMT sequences.

In this review we discuss options to minimize or even avoid NUMTs by specifically quantifying the mtDNA portion in the DNA extract of a forensic sample, and based on this knowledge, adjusting the amount of input DNA to a level that dilutes nDNA (and NUMTs) practically out and only mtDNA is remaining. This has been shown efficient for the majority of samples but can have adverse effects, such as when working with mixtures.

Additionally, we discuss post-laboratory options to filter NUMTs bioinformatically from the sequence data. It has been shown that NUMTs can be successfully reduced or even removed by either increasing the stringency of mapping parameters, so that NUMTs do not align to the reference mtDNA sequence, or by mapping the sequence data to the entire human genome including the mtDNA reference, thus allowing NUMTs to align to nDNA. Both methods have also limitations with respect to the representation (e.g., read depth) of the authentic mtDNA haplotype and minor contributions (e.g., point heteroplasmic events). Recent research has made significant progress in this respect, and future developments seem promising for improved tools to identify and remove NUMT sequences from forensic sample data more effectively. We finally describe features that can be used to discern NUMT sequences from other artefacts, such as mixture, point heteroplasmy, sequencing error or cytosine deamination. These characteristics however show that education and training are crucial to understanding and correctly interpreting MPS-generated data and reliably calling mtDNA haplotypes.

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Conflict of interest

The authors declare no conflict of interest.

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